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Agentwin

AI-Assisted Operational Digital Twin Platform for IoT & Edge Systems

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Introduction

Agentwin is a local-first, AI-assisted operational Digital Twin platform for IoT and edge systems. It connects physical devices with live telemetry, events, and verified controls, providing a real-time view of system health and operational state.

> **“Every system. Its own agent.”**



Systems
Devices and edge hardware

Live Telemetry
Real-time operational data

Digital Twin
Live system representation

Operations
Events, insights, and controls

Methodology

1. **Define the System**
Configure the project, devices, telemetry fields, operational rules, and freshness thresholds.

2. **Connect Physical Devices**
Integrate IoT and edge devices through MQTT or USB serial gateways.

3. **Validate & Store Telemetry**
Collect, timestamp, validate, and store device readings while tracking connectivity and data freshness.

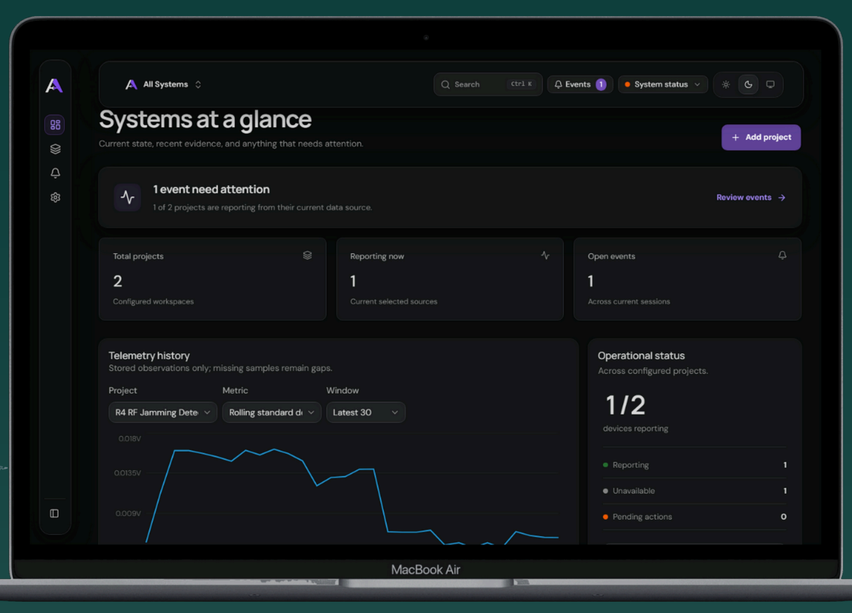
4. **Build the Operational Twin**
Transform the configured structure and live observations into a session-aware Digital Twin.

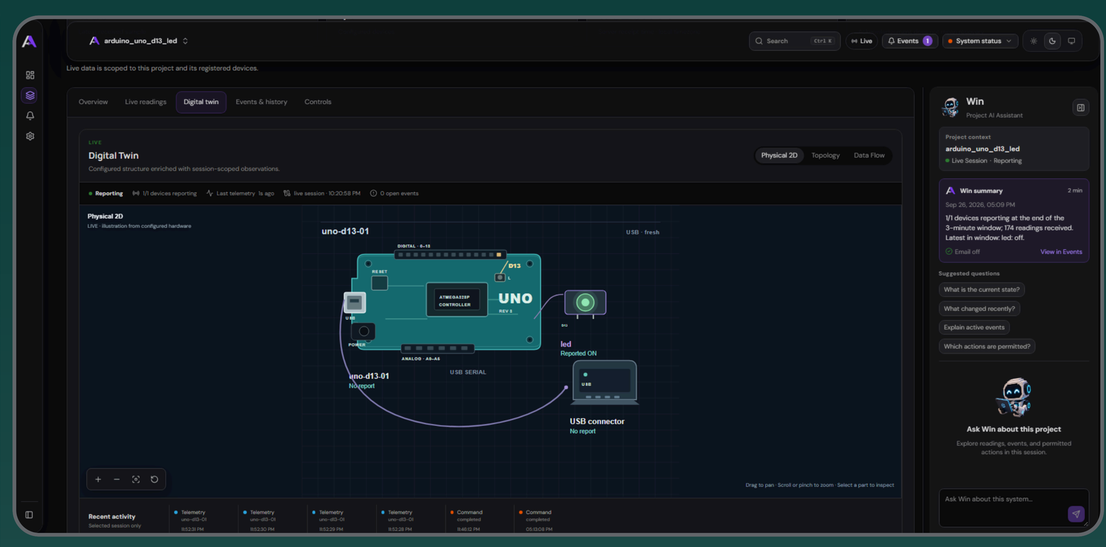
5. **Inspect Evidence with AI Assistant**
Interact with a project-scoped AI assistant to analyze data, explain events, and provide insights

6. **Support Controlled Action**
Send authorized commands and track their lifecycle until matching device telemetry confirms the physical result.

Innovation

Agentwin is designed for IoT systems, smart cities, and smart industries. It brings physical systems into operational Digital Twins by unifying live telemetry, system visualization, deterministic event detection, evidence-based AI assistance, and device control within one local platform.







Results: MVP Capabilities

- Detect and log important events automatically
- Multi-Project Support
- Unified monitoring of IoT and edge systems
- USB & Wi-Fi/MQTT integration
- Visualizes physical components, data flow, and system topology
- Local-first monitoring with private AI inference

Conclusion

Agentwin demonstrates a local-first, AI-assisted approach to operating IoT and edge systems through live Digital Twins. By unifying telemetry, system visualization, deterministic monitoring, evidence-based insights, and verified device control, the MVP provides a practical foundation for safer and more transparent connected operations.

Future Work

- Adding role-based access control, authentication, and multi-user collaboration
- 3D visualization
- Improving cybersecurity, encrypted device communication, and credential management

References & More info.

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